$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\end{pmatrix} \begin{pmatrix}2\\1\end{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}5\\4\end{pmatrix} \begin{pmatrix}6\\5\end{pmatrix} \begin{pmatrix}6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Initial $W = \begin{vmatrix}4\\-2\\-2\end{vmatrix}$ **Step η = 2**

C1 *positive,* **C2** *negative,* **Zero** is unaccepted

Check classes of $\underset{Z_1}{\begin{pmatrix}0\\0\end{pmatrix}} \underset{Z_1}{\begin{pmatrix}7\\6\end{pmatrix}} \underset{Z_1}{\begin{pmatrix}3\\2\end{pmatrix}}$

Ugmented Vectors (from Top)

x_1	x_2	x_3	x_4	x_5	x_6
(1)	(1)	(1)	(1)	(1)	(1)
(1)	(2)	(1)	(5)	(6)	(6)
(1)	(2)	(4)	(5)	(6)	(6)
C_1	C_1	C_1	C_2	C_2	C_2

Iteration 3

$d(x_1) = \begin{pmatrix}1 & 1 & 1\end{pmatrix}\begin{pmatrix}8\\-12\\-8\end{pmatrix} = -12$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}8\\-12\\-8\end{pmatrix} + 2 * \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}10\\-10\\-6\end{pmatrix}$ ⇐

$d(x_2) = \begin{pmatrix}1 & 2 & 1\end{pmatrix}\begin{pmatrix}10\\-10\\-6\end{pmatrix} = -16$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}10\\-10\\-6\end{pmatrix} + 2 * \begin{pmatrix}1\\2\\1\end{pmatrix} = \begin{pmatrix}12\\-6\\-4\end{pmatrix}$ ⇐

$d(x_3) = \begin{pmatrix}1 & 1 & 2\end{pmatrix}\begin{pmatrix}12\\-6\\-4\end{pmatrix} = -2$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}12\\-6\\-4\end{pmatrix} + 2 * \begin{pmatrix}1\\1\\2\end{pmatrix} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$ ⇐

$d(x_4) = \begin{pmatrix}1 & 5 & 4\end{pmatrix}\begin{pmatrix}14\\-4\\0\end{pmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

$d(x_5) = \begin{pmatrix}1 & 6 & 5\end{pmatrix}\begin{pmatrix}14\\-4\\0\end{pmatrix} = -10$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

$d(x_6) = \begin{pmatrix}1 & 6 & 6\end{pmatrix}\begin{pmatrix}14\\-4\\0\end{pmatrix} = -112$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$